

CHAPTER 03

Activities & Lab Packet

Reproduction & Heredity

A complete set of labs, worksheets, writing prompts, discussions, and a group project to accompany Chapter 3. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S7L3

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1 Coin-Toss Punnett Lab

LAB

OBJECTIVE

Use coin flips to model how alleles combine and compare results to Punnett-square predictions.

TIME

30–40 min

STANDARDS

S7L3

Materials (per pair)

- 2 coins (one per "parent"). Heads = T (dominant), Tails = t (recessive)
- Tally sheet (next page)

Student Instructions

1. Each partner is a Tt parent. Flip both coins together — that's one "offspring."
2. Record the combo (TT, Tt, or tt) and whether it can roll its tongue (TT/Tt = yes, tt = no).
3. Repeat for 20 offspring. Tally your results.
4. Build the Tt × Tt Punnett square and compare your actual % to the predicted 75% / 25%.

! TEACHER NOTES

The "aha": small samples wander from the prediction; pooling the whole class's data gets very close to 75/25. Use this to teach **probability** vs. a single outcome.

Coin-Toss — Tally Sheet

Name: _____ Period: _____

Genotype	Tally (20 flips)	Count	My %	Predicted %
TT				25%
Tt				50%
tt				25%

How close were your results to the prediction? Why might they differ?

2 Trait Inheritance — Build a Critter

LAB

OBJECTIVE

Use two parents' alleles to determine and draw the traits of their offspring.

TIME

40 min

STANDARDS

S7L3

Materials (per pair)

- Trait/allele key (below)
- 2 bags of allele slips, OR coins (Heads = dominant, Tails = recessive)
- Paper & markers to draw the critter

Trait Key

Trait	Dominant (capital)	Recessive (lower)
Body color	G = green	g = yellow
Antennae	A = present	a = none
Spots	S = spotted	s = plain
Tail	T = curly	t = straight

Student Instructions

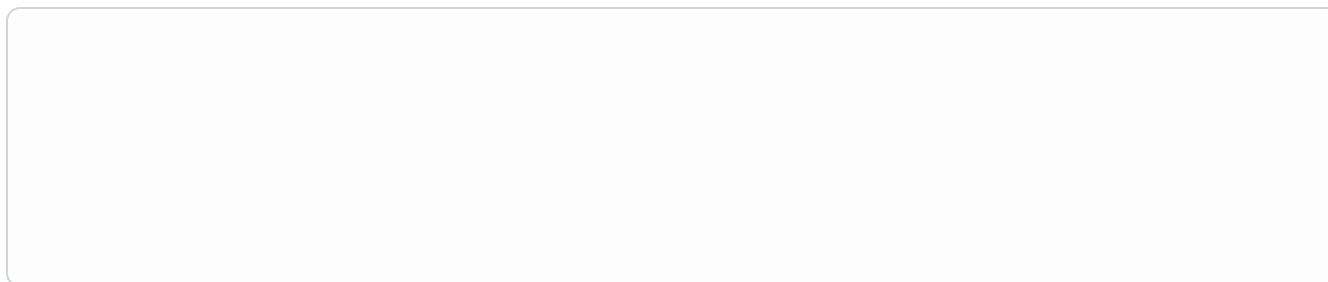
1. For each trait, draw one allele from "Parent 1" and one from "Parent 2" to make the offspring's genotype.
2. Use dominance rules to find the phenotype (the trait you'd see).
3. Record genotype + phenotype, then draw your critter.

Build-a-Critter — Data Sheet

Name: _____ Period: _____

Trait	Genotype (e.g., Gg)	Phenotype (what you see)
Body color		
Antennae		
Spots		
Tail		

Draw your critter:



3 Asexual vs. Sexual Modeling

LAB

OBJECTIVE

Model both kinds of reproduction and explain why one creates variation.

TIME

30 min

STANDARDS

S7L3

Student Instructions

1. Asexual: take a "parent" strip of 6 colored beads. Copy it exactly twice — those are the clones.
2. Sexual: take HALF the beads from two different parents and combine them into one offspring.
3. Compare: which method made identical offspring? Which made a unique mix?

Reproduction Modeling — Data Sheet

Name: _____ Period: _____

Feature	Asexual	Sexual
Number of parents		
Offspring DNA (same/mixed)		
Genetic variation		
Example organism		

Why is genetic variation like "not putting all your eggs in one basket"?

ANSWER KEY

Asexual: one parent, identical DNA, no variation (bacteria, hydra, starfish). Sexual: two parents, mixed DNA, lots of variation. Variation helps a species survive new diseases or changes.

4 Genes & Chromosomes

WORKSHEET

OBJECTIVE

Order the structures of heredity and define each term.

TIME

15 min

STANDARDS

S7L3

Name: _____ Period: _____

Part A: Order from largest to smallest (1 = biggest container).

Gene ____ Cell ____ DNA ____ Chromosome ____

Part B: Match the term to its meaning.

Term	Meaning
DNA	
Gene	
Chromosome	

Part C: Short answer.

- How many chromosomes are in a human body cell, and how many came from each parent?

- The textbook compares DNA structures to nesting dolls. Why is that a good comparison? _____

ANSWER KEY

Part A: Cell 1, Chromosome 2, Gene 3, DNA 4. Part B: DNA = chemical code with all instructions; Gene = short DNA section controlling one trait; Chromosome = long coiled DNA strand holding genes. Part C: 1) 46 (23 pairs), 23 from each parent; 2) each structure fits inside the larger one (cell > chromosome > gene > DNA).

5 Punnett Square Practice

WORKSHEET

OBJECTIVE

Build Punnett squares and find the probability of offspring traits.

TIME

20 min

STANDARDS

S7L3

Name: _____ Period: _____

For tongue rolling, **T** = can roll (dominant), **t** = cannot (recessive). Fill each square and answer.

1) Cross $Tt \times Tt$

	T	t
T		
t		

% that can roll: _____ % that cannot: _____

2) Cross $TT \times tt$

	t	t
T		
T		

% that can roll: _____ Genotype of all offspring: _____

3) Cross $Tt \times tt$

	t	t
T		
t		

% that can roll: _____ % that cannot: _____

ANSWER KEY

1) $TT, Tt, Tt, tt \rightarrow 75\%$ roll, 25% cannot. 2) all $Tt \rightarrow 100\%$ roll, genotype Tt . 3) $Tt, Tt, tt, tt \rightarrow 50\%$ roll, 50% cannot.

6 Genotype & Phenotype

WORKSHEET

OBJECTIVE

Tell genotype from phenotype and identify homozygous vs. heterozygous.

TIME

15 min

STANDARDS

S7L3

Name: _____ Period: _____

Part A: For each genotype, write the name (homozygous dominant / heterozygous / homozygous recessive).

Genotype	Name	Genotype	Name
BB		Tt	
bb		RR	
Gg		ww	

Part B: B = brown eyes (dominant), b = blue eyes. Write the phenotype.

BB → _____ Bb → _____ bb → _____

Part C: Short answer.

1. Which letter combination is the only way a recessive trait can show up? _____
2. What is the difference between genotype and phenotype? _____

ANSWER KEY

Part A: BB hom. dominant; bb hom. recessive; Gg heterozygous; Tt heterozygous; RR hom. dominant; ww hom. recessive. Part B: BB brown, Bb brown, bb blue. Part C: 1) two lowercase (bb); 2) genotype = the gene combo, phenotype = the visible trait.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain heredity in writing.

TIME

15–30 min each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Born or Earned?** Explain the difference between an inherited trait and an acquired trait, with one example of each from your own life.
2. **Why You Look Like Family.** Use the words gene, DNA, and heredity to explain why kids often look like their parents.

Level 2 — Intermediate (3 paragraphs)

3. **Two Ways to Reproduce.** Compare asexual and sexual reproduction. Explain the trade-off between speed and variation, and when each would be an advantage.
4. **The Punnett Detective.** Two brown-eyed parents have a blue-eyed child. Use alleles and a Punnett square to explain how this is possible.

Level 3 — Advanced (essay)

5. **Designing Life.** Humans used selective breeding to turn one wild mustard plant into broccoli, kale, and cabbage. Construct an argument about whether selective breeding is helpful, risky, or both — using specific examples.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific reasoning through think-pair-share.

TIME

Warm-ups, exit tickets, seminar

STANDARDS

Cross-cutting

TEACHER NOTES

Think-pair-share format. Several questions have no single right answer — push for evidence from the chapter.

Quick Warm-Ups (1–2 min)

1. If you get a scar, will your future children have it? Why or why not?
2. Can two parents who CAN roll their tongues have a child who can't?
3. Why are identical twins so much alike?
4. Is a banana plant grown from a cutting the parent's clone or its child?
5. Why do farmers pick the biggest cows to breed?

Deeper Discussions (5–15 min)

6. Why does sexual reproduction help a species survive a new disease better than asexual reproduction?
7. If a Punnett square says 25% chance of a trait, does that mean exactly 1 of every 4 offspring will have it? Explain.
8. Selective breeding and natural selection both change traits over time. What is the key difference?
9. You share 99.9% of your DNA with every human. Why does the tiny 0.1% matter so much?
10. Should humans use selective breeding (or genetic tools) on crops and animals? What are the benefits and risks?

9 Selective Breeding Design Challenge

GROUP

OBJECTIVE

Design a selective-breeding plan to develop a useful new variety.

TIME

2 class periods

STANDARDS

S7L3

Project Description

Each group acts as a team of breeders. They pick a starting organism and a goal trait, then design a multi-generation breeding plan and present it as a poster or pitch.

Plan Requirements

The Goal

- Starting organism (real or invented)
- The desired trait & why it's useful
- Whether the trait is dominant or recessive

The Plan

- Which individuals you'd breed each generation, and why
- A Punnett square showing one cross
- One risk of selective breeding for your variety

Presentation Day

Each group pitches their variety in 2–3 minutes. The class votes on which plan is most likely to succeed and explains why.

TEACHER NOTES

Idea starters: drought-resistant corn, friendlier service dog, a flower with a new color, fast-growing tomato. **Grading:** sound genetics 40%, clear plan 25%, creativity 15%, presentation 20%.

10 Vocabulary Bingo

REVIEW

OBJECTIVE

Review chapter vocabulary in a low-stress game.

TIME

20–30 min

STANDARDS

S7L3

Setup

Blank 5×5 grid; word list on board; FREE middle. Read a clue, students mark the word. 5 in a row wins.

Vocabulary List (24 words)

trait	heredity	DNA	gene	chromosome
allele	dominant	recessive	genotype	phenotype
homozygous	heterozygous	Punnett square	probability	clone
asexual reproduction	sexual reproduction	genetic variation	mutation	selective breeding
offspring	inherited	acquired	nucleus	—

Sample Clue Cards (read aloud, don't show)

Clue to read aloud	Answer
The passing of traits from parents to offspring	heredity
A short section of DNA that controls one trait	gene
A long, coiled strand of DNA	chromosome
The stronger allele, shown with a capital letter	dominant
The weaker allele, shown with a lowercase letter	recessive
The gene combination an organism has (like Tt)	genotype
The physical trait you can see	phenotype
Having two different alleles (Tt)	heterozygous
A tool that predicts offspring trait chances	Punnett square
Reproduction with one parent making clones	asexual reproduction
A change in DNA that can create a new trait	mutation
When humans choose which organisms reproduce	selective breeding

⚠️ TEACHER NOTES

Easy adaptation: give the list with definitions during play. Run multiple patterns so several students win.